Before converting Eq. (1.71) into the phasor domain, we express Eq. (1.70) in terms of a cosine reference:

$$\begin{aligned} v_s(t) &= 5\sin(4\times10^4 t - 30^\circ) \\ &= 5\cos(4\times10^4 t - 120^\circ) \quad \text{(V)}. \end{aligned} \tag{1.72}$$

The coefficient of t specifies the angular frequency as $\omega = 4\times10^4$ (rad/s). Per the second entry in **Table 1-5**, the voltage phasor corresponding to $v_s(t)$ is

$$\tilde{V}_s = 5e^{-j120^\circ} \quad \text{(V)},$$

and the phasor equation corresponding to Eq. (1.71) is

$$R\tilde{I} + j\omega L\tilde{I} = \tilde{V}_s. \tag{1.73}$$

Solving for the current phasor $\tilde{I}$, we have

$$\begin{aligned} \tilde{I} &= \frac{\tilde{V}_s}{R + j\omega L} \\ &= \frac{5e^{-j120^\circ}}{6 + j4\times10^4\times2\times10^{-4}} = \frac{5e^{-j120^\circ}}{10e^{j53.1^\circ}} = 0.5e^{-j173.1^\circ} \quad \text{(A)}. \end{aligned}$$

The voltage phasor across the inductor is related to $\tilde{I}$ by

$$\begin{aligned} \tilde{V}_L &= j\omega L\tilde{I} \\ &= j4\times10^4\times2\times10^{-4}\times0.5e^{-j173.1^\circ} \\ &= 4e^{j(90^\circ-173.1^\circ)} = 4e^{-j83.1^\circ} \quad \text{(V)}, \end{aligned}$$

and the corresponding instantaneous voltage $v_L(t)$ is therefore

$$\begin{aligned} v_L(t) &= \Re\mathfrak{e}\left[\tilde{V}_L e^{j\omega t}\right] \\ &= \Re\mathfrak{e}\left[4e^{-j83.1^\circ}e^{j4\times10^4 t}\right] \\ &= 4\cos(4\times10^4 t - 83.1^\circ) \quad \text{(V)}. \end{aligned}$$

Concept Question 1-12: Why is the phasor technique useful? When is it used? Describe the process.

Concept Question 1-13: How is the phasor technique used when the forcing function is a nonsinusoidal periodic waveform, such as a train of pulses?

Exercise 1-11: A series RL circuit is connected to a voltage source given by $v_s(t) = 150\cos\omega t$ (V). Find (a) the phasor current $\tilde{I}$ and (b) the instantaneous current $i(t)$ for $R = 400\ \Omega$, $L = 3$ mH, and $\omega = 10^5$ rad/s.

Answer: (a) $\tilde{I} = 150/(R + j\omega L) = 0.3\angle{-36.9^\circ}$ (A), (b) $i(t) = 0.3\cos(\omega t - 36.9^\circ)$ (A). (See EM.)

Exercise 1-12: A phasor voltage is given by $\tilde{V} = j5$ V. Find $v(t)$.

Answer: $v(t) = 5\cos(\omega t + \pi/2) = -5\sin\omega t$ (V). (See EM.)

1-7.2 Traveling Waves in the Phasor Domain

According to **Table 1-5**, if we set $\phi_0 = 0$, its third entry becomes

$$A\cos(\omega t + \beta x) \quad \Longleftrightarrow \quad Ae^{j\beta x}. \tag{1.74}$$

From the discussion associated with Eq. (1.31), we concluded that $A\cos(\omega t + \beta x)$ describes a wave traveling in the negative x direction.

▶ In the phasor domain, a wave of amplitude A traveling in the positive x direction in a lossless medium with a phase constant β is given by the negative exponential $Ae^{-j\beta x}$. Conversely, a wave traveling in the negative x direction is given by $Ae^{j\beta x}$. Thus, the sign of x in the exponential is opposite to the direction of travel. ◀

例1-4续解与相量域中的行波

先将式（1.70）的电源电压改写为余弦参考，再把式（1.71）转换为相量方程：

$$\begin{aligned} v_s(t) &= 5\sin(4\times10^4 t - 30^\circ) \\ &= 5\cos(4\times10^4 t - 120^\circ) \quad \text{(V)}. \end{aligned} \tag{1.72}$$

$$R\tilde{I} + j\omega L\tilde{I} = \tilde{V}_s. \tag{1.73}$$

角频率ω = 4 × 10^4 rad/s。由表1-5，电源相量为5∠(−120°) V；回路阻抗R + jωL = 6 + j8 = 10∠53.1° Ω。求电流相量：

$$\begin{aligned} \tilde{I} &= \frac{\tilde{V}_s}{R + j\omega L} \\ &= \frac{5e^{-j120^\circ}}{6 + j4\times10^4\times2\times10^{-4}} \\ &= \frac{5e^{-j120^\circ}}{6 + j8} = \frac{5e^{-j120^\circ}}{10e^{j53.1^\circ}} = 0.5e^{-j173.1^\circ} \quad \text{(A)}. \end{aligned}$$

电感电压相量等于电流相量乘jωL；再乘$e^{j\omega t}$并取实部，即得瞬时电压：

$$\begin{aligned} \tilde{V}_L &= j\omega L\tilde{I} \\ &= j4\times10^4\times2\times10^{-4}\times0.5e^{-j173.1^\circ} \\ &= 4e^{j(90^\circ-173.1^\circ)} = 4e^{-j83.1^\circ} \quad \text{(V)}, \end{aligned}$$

电感电压相量。

$$\begin{aligned} v_L(t) &= \Re\mathfrak{e}\left[\tilde{V}_L e^{j\omega t}\right] \\ &= \Re\mathfrak{e}\left[4e^{-j83.1^\circ}e^{j4\times10^4 t}\right] \\ &= 4\cos(4\times10^4 t - 83.1^\circ) \quad \text{(V)}. \end{aligned}$$

对应的瞬时电压。

概念题1-12：相量法为什么有用？适用于何种情况？说明求解过程。
概念题1-13：激励为脉冲串等非正弦周期波形时，如何使用相量法？

练习1-11：串联RL电路的电源电压为v_s(t) = 150 cosωt V。已知R = 400 Ω，L = 3 mH，ω = 10^5 rad/s，求（a）电流相量；（b）瞬时电流。
答案：（a）150/(400 + j300) = 0.3∠(−36.9°) A；（b）i(t) = 0.3 cos(ωt − 36.9°) A。详见配套资源。

练习1-12：电压相量为j5 V，求瞬时电压。
答案：v(t) = 5 cos(ωt + π/2) = −5 sinωt V。详见配套资源。

1-7.2 相量域中的行波

在表1-5第三项中令φ_0 = 0，得到：

$$A\cos(\omega t + \beta x) \quad \Longleftrightarrow \quad Ae^{j\beta x}. \tag{1.74}$$

由式（1.31）的讨论，A cos(ωt + βx)沿−x方向传播。在本书采用的$e^{j\omega t}$时间因子、β > 0约定下，无损介质中沿+x方向传播的波，其相量为$Ae^{-j\beta x}$；沿−x方向传播的波，其相量为$Ae^{j\beta x}$。空间指数中x项的符号与传播方向相反。

校核：例1-4电感电压振幅为4 V，余弦参考相位约−83.13°；练习1-11电流振幅0.3 A、相位约−36.87°。上述相位取到原书精度；相量法结果为正弦稳态解。

Chapter 1 Summary

Concepts

- Electromagnetics is the study of electric and magnetic phenomena and their engineering applications.
- The International System of Units consists of the six fundamental dimensions listed in **Table 1-1**. The units of all other physical quantities can be expressed in terms of the six fundamental units.
- The four fundamental forces of nature are nuclear, weak-interaction, electromagnetic, and gravitational forces.
- The source of the electric field quantities **E** and **D** is the electric charge q. In a material, **E** and **D** are related by $\mathbf{D} = \varepsilon\mathbf{E}$, where ε is the electrical permittivity of the material. In free space, $\varepsilon = \varepsilon_0 \approx (1/36\pi) \times 10^{-9}$ (F/m).
- The source of the magnetic field quantities **B** and **H** is the electric current I. In a material, **B** and **H** are related by $\mathbf{B} = \mu\mathbf{H}$, where μ is the magnetic permeability of the medium. In free space, $\mu = \mu_0 = 4\pi \times 10^{-7}$ (H/m).
- Electromagnetics consists of three branches:
  (1) electrostatics, which pertains to stationary charges,
  (2) magnetostatics, which pertains to dc currents, and
  (3) electrodynamics, which pertains to time-varying currents.
- A traveling wave is characterized by a spatial wavelength λ, a time period T, and a phase velocity $u_p = \lambda/T$.
- An electromagnetic (EM) wave consists of oscillating electric and magnetic field intensities and travels in free space at the velocity of light: $c = 1/\sqrt{\varepsilon_0\mu_0}$. The EM spectrum encompasses gamma rays, X-rays, visible light, infrared waves, and radio waves.
- Phasor analysis is a useful mathematical tool for solving problems involving time-periodic sources.

Mathematical and Physical Models

Electric field due to charge q in free space

$$\mathbf{E} = \hat{\mathbf{R}}\frac{q}{4\pi\varepsilon_0 R^2}$$

Magnetic field due to current I in free space

$$\mathbf{B} = \hat{\boldsymbol{\phi}}\frac{\mu_0 I}{2\pi r}$$

Plane wave $y(x,t) = Ae^{-\alpha x}\cos(\omega t - \beta x + \phi_0)$

- $\alpha = 0$ in lossless medium
- phase velocity $u_p = f\lambda = \frac{\omega}{\beta}$
- $\omega = 2\pi f$; $\beta = 2\pi/\lambda$
- ϕ_0 = phase reference

Complex numbers

- Euler's identity
  $e^{j\theta} = \cos\theta + j\sin\theta$
- Rectangular-polar relations
  $x = |z|\cos\theta, \quad y = |z|\sin\theta,$
  $|z| = \sqrt[+]{x^2+y^2}, \quad \theta = \tan^{-1}(y/x)$

Phasor-domain equivalents

See **Table 1-5**

PROBLEMS

Section 1-4: Traveling Waves

*1.1 A harmonic wave traveling along a string is generated by an oscillator that completes 180 vibrations per minute. If it is observed that a given crest, or maximum, travels 300 cm in 10 s, what is the wavelength?

1.2 For the pressure wave described in Example 1-1, plot the following:

(a) $p(x,t)$ versus x at $t = 0$

(b) $p(x,t)$ versus t at $x = 0$

Be sure to use appropriate scales for x and t so that each of your plots covers at least two cycles.

*1.3 A 2 kHz sound wave traveling in the x direction in air was observed to have a differential pressure $p(x,t) = 10$ N/m^2 at $x = 0$ and $t = 50$ μs. If the reference phase of $p(x,t)$ is 36°, find a complete expression for $p(x,t)$. The velocity of sound in air is 330 m/s.

* Answer(s) available in Appendix E.

第一章总结

概念

- 电磁学研究电、磁现象及其工程应用。
- 国际单位制（SI）有七个基本单位。表1-1列出本书常用的六种基本量，未列物质的量及其单位摩尔。[校注1]
- 自然界的四种基本相互作用为强相互作用、弱相互作用、电磁相互作用和引力相互作用。
- 电荷 q 可产生电场。在本书采用的线性各向同性介质中，D=εE，ε为介电率；真空中 $\varepsilon_0 \approx (1/36\pi)\times 10^{-9}$ F/m。
- 电流 I 可产生磁场；在线性各向同性介质中，B=μH，μ为磁导率。真空中 $\mu_0 \approx 4\pi\times 10^{-7}$ H/m。[校注1]
- 本书将电磁学分为静电学（静止电荷）、静磁学（恒定电流）和电动力学（时变电磁现象）。
- 行波由空间波长 λ、时间周期 T 和相速度 $u_p=\lambda/T$ 描述。
- 电磁波由随时间变化的电场与磁场组成，在真空中以光速 $c=1/\sqrt{(\varepsilon_0\mu_0)}$ 传播。电磁频谱包括伽马射线、X射线、紫外线、可见光、红外线、微波和无线电波等。
- 相量分析适用于同一频率的正弦稳态问题。非正弦周期信号可先作傅里叶分解，再分别分析各频率分量。

数学与物理模型

真空中点电荷 q 产生的电场

$$\mathbf{E} = \hat{\mathbf{R}}\frac{q}{4\pi\varepsilon_0 R^2}$$

真空中无限长直导线电流 I 产生的磁场

$$\mathbf{B} = \hat{\boldsymbol{\phi}}\frac{\mu_0 I}{2\pi r}$$

平面波 $y(x,t) = Ae^{-\alpha x}\cos(\omega t - \beta x + \phi_0)$

- $\alpha = 0$ 在无损介质中
- 相速度 $u_p = f\lambda = \frac{\omega}{\beta}$
- $\omega = 2\pi f$; $\beta = 2\pi/\lambda$
- ϕ_0 = 参考相位

复数

- 欧拉恒等式
  $e^{j\theta} = \cos\theta + j\sin\theta$
- 直角形式与极坐标形式的关系
  $x = |z|\cos\theta, \quad y = |z|\sin\theta,$
  $|z| = \sqrt[+]{x^2+y^2}, \quad \theta = \tan^{-1}(y/x)$

相量域对应关系

见表1-5

辐角 θ 须结合 x、y 的象限确定；z=0 时辐角无定义。

问题

第 1-4 节：行波

*1.1 沿弦传播的谐波由每分钟完成 180 次振动的振荡器产生。如果观察到给定的波峰或最大值在 10 秒内传播 300 厘米，那么波长是多少？

1.2 对于例 1-1 中描述的压力波，绘制下列的：

(a) p(x,t) 与 t = 0 处的 x

(b) p(x,t) 与 x = 0 处的 t

请务必对 x 和 t 使用适当的比例，以便每个图至少涵盖两个周期。

*1.3 观察到在空气中沿 x 方向传播的 2 kHz 声波在 x = 0 和 t = 50 μ s 处具有压差 p(x,t) = 10 N/m^2。如果p(x,t)的参考相位是36°，则找到p(x,t)的完整表达式。空气中的声速为 330 m/s。

* 标记的习题在附录 E 提供答案。

校注1：原书“六个基本单位”不完整。另自2019年SI修订后，μ_0不再由定义固定为$4\pi\times10^{-7}$ H/m；此处为工程近似。依据：BIPM《SI手册》第9版及第26届CGPM第1号决议。

Important Terms

Provide definitions or explain the meaning of the following terms:

angular velocity ω
attenuation constant α
attenuation factor
Biot–Savart law
complex conjugate
complex number
conductivity σ
constitutive parameters
continuous periodic wave
Coulomb's law
dielectric constant
dynamic
electric dipole
electric field intensity **E**
electric flux density **D**
electric polarization
electrical force
electrical permittivity ε
electrodynamics
electrostatics
EM spectrum
Euler's identity
forcing function
fundamental dimensions
instantaneous function
law of conservation of electric charge
LCD
liquid crystal
lossless or lossy medium
magnetic field intensity **H**
magnetic flux density **B**
magnetic force
magnetic permeability μ
magnetostatics
microwave band
monochromatic
nonmagnetic materials
perfect conductor
perfect dielectric
periodic
phase
phase constant (wave number) β
phase lag and lead
phase velocity (propagation velocity) u_p
phasor
plane wave
principle of linear superposition
reference phase ϕ_0
relative permittivity or dielectric constant ε_r
SI system of units
static
transient wave
velocity of light c
wave amplitude
wave frequency f
wave period T
waveform
wavelength λ

1.4 A wave traveling along a string is given by

$$y(x,t) = 2\sin(4\pi t + 10\pi x) \qquad \text{(cm)},$$

where x is the distance along the string in meters and y is the vertical displacement. Determine: (a) the direction of wave travel, (b) the reference phase ϕ_0, (c) the frequency, (d) the wavelength, and (e) the phase velocity.

1.5 Two waves, $y_1(t)$ and $y_2(t)$, have identical amplitudes and oscillate at the same frequency, but $y_2(t)$ leads $y_1(t)$ by a phase angle of 60°. If

$$y_1(t) = 4\cos(2\pi \times 10^3 t),$$

write the expression appropriate for $y_2(t)$ and plot both functions over the time span from 0 to 2 ms.

*1.6 The height of an ocean wave is described by the function

$$y(x,t) = 1.5\sin(0.5t - 0.6x) \qquad \text{(m)}.$$

Determine the phase velocity and wavelength, and then sketch $y(x,t)$ at $t = 2s$ over the range from $x = 0$ to $x = 2\lambda$.

1.7 A wave traveling along a string in the $+x$ direction is given by

$$y_1(x,t) = A\cos(\omega t - \beta x),$$

where $x = 0$ is the end of the string, which is tied rigidly to a wall, as shown in **Fig. P1.7**.

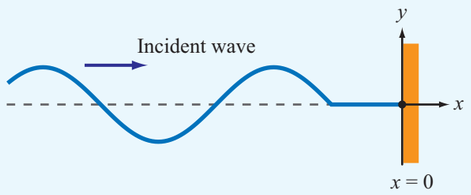


Figure P1.7 Wave on a string tied to a wall at $x = 0$ (Problem 1.7).

When wave $y_1(x,t)$ arrives at the wall, a reflected wave $y_2(x,t)$ is generated. Hence, at any location on the string, the vertical displacement y_s is the sum of the incident and reflected waves:

$$y_s(x,t) = y_1(x,t) + y_2(x,t).$$

(a) Write an expression for $y_2(x,t)$, keeping in mind its direction of travel and the fact that the end of the string cannot move.

(b) Generate plots of $y_1(x,t)$, $y_2(x,t)$, and $y_s(x,t)$ versus x over the range $-2\lambda \le x \le 0$ at $\omega t = \pi/4$ and at $\omega t = \pi/2$.

重要术语 请给出下列术语的定义或解释其含义：

角频率 ω
衰减常数 α
衰减因子
毕奥–萨伐尔定律
复共轭
复数
电导率 σ
本构参数
连续周期波
库仑定律
介电常数
时变的
电偶极子
电场强度 E
电通量密度 D
电极化
电场力
介电率 ε
电动力学
静电学
电磁频谱
欧拉恒等式
激励函数
基本量纲
瞬时函数
电荷守恒定律
液晶显示器（LCD）
液晶
无损介质或有损介质
磁场强度 H
磁通密度 B
磁力
磁导率 μ
静磁学
微波波段
单色的
非磁性材料
理想导体
理想电介质
周期的
相位
相位常数（波数）β
相位滞后与超前
相速度（传播速度）u_p
相量
平面波
线性叠加原理
参考相位 φ_0
相对介电率 ε_r（介电常数）
国际单位制（SI）
静态的
瞬态波
光速 c
波的振幅
波的频率 f
波的周期 T
波形
波长 λ

1.4 沿弦传播的波为

$$y(x,t) = 2\sin(4\pi t + 10\pi x) \qquad \text{(cm)},$$

其中，x 为沿弦的距离（单位为 m），y 为竖直位移。求：(a) 波的传播方向；(b) 参考相位 φ_0；(c) 频率；(d) 波长；(e) 相速度。

1.5 两个波 $y_1(t)$ 与 $y_2(t)$ 的振幅、频率相同，$y_2(t)$ 的相位比 $y_1(t)$ 超前 60°。若

$$y_1(t) = 4\cos(2\pi \times 10^3 t),$$

写出 $y_2(t)$ 的表达式，并绘制 0 至 2 ms 内这两个函数的曲线。

*1.6 海浪的高度由下式描述：

$$y(x,t) = 1.5\sin(0.5t - 0.6x) \qquad \text{(m)}.$$

求相速度及波长，并画出 t = 2 s 时在 0 ≤ x ≤ 2λ 范围内的波形 y(x,t)。

1.7 沿弦向 +x 方向传播的波为

$$y_1(x,t) = A\cos(\omega t - \beta x),$$

其中，x = 0 为弦的固定端，该端牢固地固定在墙上，如图 P1.7 所示。

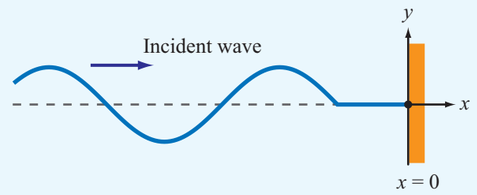


图 P1.7 弦在 x = 0 处固定于墙上时的波（习题 1.7）。图中箭头表示入射波。

当 $y_1(x,t)$ 到达墙壁时，将产生反射波 $y_2(x,t)$。因此，弦上任意位置的竖直位移 y_s 为入射波与反射波的位移之和：

$$y_s(x,t) = y_1(x,t) + y_2(x,t).$$

(a) 写出 $y_2(x,t)$ 的表达式，注意其传播方向以及弦的固定端不能移动这一条件。

(b) 分别在 ωt = π/4 和 ωt = π/2 时，绘制 −2λ ≤ x ≤ 0 范围内 $y_1(x,t)$、$y_2(x,t)$ 与 $y_s(x,t)$ 随 x 变化的曲线。

校注：术语表中的 angular velocity 在本章波动语境下指角频率 ω，按该物理含义译出。

1.8 Two waves on a string are given by the functions:

$$y_1(x,t) = 4\cos(20t - 30x) \qquad \text{(cm)}$$
$$y_2(x,t) = -4\cos(20t + 30x) \qquad \text{(cm)},$$

where x is in centimeters. The waves are said to interfere constructively when their superposition $|y_s| = |y_1 + y_2|$ is a maximum, and they interfere destructively when $|y_s|$ is a minimum.

*(a) What are the directions of propagation of waves $y_1(x,t)$ and $y_2(x,t)$?

(b) At $t = (\pi/50)$ s, what location x do the two waves interfere constructively, and what is the corresponding value of $|y_s|$?

(c) At $t = (\pi/50)$ s, what location x do the two waves interfere destructively, and what is the corresponding value of $|y_s|$?

1.9 Find expressions for $y(x,t)$ for a sinusoidal wave traveling along a string in the negative x direction, given that $y_{max} = 40$ cm, $\lambda = 30$ cm, $f = 10$ Hz, and

(a) $y(x,0) = 0$ at $x = 0$

(b) $y(x,0) = 0$ at $x = 3.75$ cm

*1.10 An oscillator that generates a sinusoidal wave on a string completes 20 vibrations in 50 s. The wave peak is observed to travel a distance of 2.8 m along the string in 5 s. What is the wavelength?

1.11 The vertical displacement of a string is given by the harmonic function:

$$y(x,t) = 2\cos(16\pi t - 20\pi x) \qquad \text{(m)},$$

where x is the horizontal distance along the string in meters. Suppose a tiny particle were attached to the string at $x = 5$ cm. Obtain an expression for the vertical velocity of the particle as a function of time.

*1.12 Given two waves characterized by

$$y_1(t) = 3\cos\omega t,$$
$$y_2(t) = 3\sin(\omega t + 60^\circ),$$

does $y_2(t)$ lead or lag $y_1(t)$ and by what phase angle?

1.13 The voltage of an electromagnetic wave traveling on a transmission line is given by

$$v(z,t) = 5e^{-\alpha z}\sin(4\pi \times 10^9 t - 20\pi z) \qquad \text{(V)},$$

where z is the distance in meters from the generator.

(a) Find the frequency, wavelength, and phase velocity of the wave.

(b) At $z = 2$ m, the amplitude of the wave was measured to be 2 V. Find α.

*1.14 A certain electromagnetic wave traveling in seawater was observed to have an amplitude of 98.02 (V/m) at a depth of 10 m, and an amplitude of 81.87 (V/m) at a depth of 100 m. What is the attenuation constant of seawater?

1.15 A laser beam traveling through fog was observed to have an intensity of 1 (μW/m^2) at a distance of 2 m from the laser gun and an intensity of 0.2 (μW/m^2) at a distance of 3 m. Given that the intensity of an electromagnetic wave is proportional to the square of its electric-field amplitude, find the attenuation constant α of fog.

1.16 In Module 1.2, set $f = 1$ Hz, $\lambda = 2$ cm, and $\alpha = 0.5$ Np/cm. Estimate the amplitude $y(x)$ of the blue wave at its first maximum (at $x = 0$) and at its second maximum (at $x = 2$ cm). Compute the ratio $y(2\text{ cm})/y(0)$. Does the result match your expectations given the values of f, λ, and α?

Section 1-5: Complex Numbers

1.17 Complex numbers z_1 and z_2 are given as

$$z_1 = 3 - j2,$$
$$z_2 = -4 + j3.$$

(a) Express z_1 and z_2 in polar form.

(b) Find $|z_1|$ by first applying Eq. (1.41) and then by applying Eq. (1.43).

*(c) Determine the product $z_1 z_2$ in polar form.

(d) Determine the ratio z_1/z_2 in polar form.

(e) Determine z_1^3 in polar form.

1.18 Evaluate each of the following complex numbers and express the result in rectangular form:

(a) $z_1 = 8e^{j\pi/3}$

*(b) $z_2 = \sqrt{3}\, e^{j3\pi/4}$

(c) $z_3 = 2e^{-j\pi/2}$

(d) $z_4 = j^3$

(e) $z_5 = j^{-4}$

(f) $z_6 = (1-j)^3$

(g) $z_7 = (1-j)^{1/2}$

1.8 弦上的两个波由以下函数给出：

$$y_1(x,t) = 4\cos(20t - 30x) \qquad \text{(cm)}$$
$$y_2(x,t) = -4\cos(20t + 30x) \qquad \text{(cm)},$$

其中 x 的单位是厘米。当波的叠加 $|y_s| = |y_1 + y_2|$ 为最大值时，波被称为相长干涉；当 $|y_s|$ 为最小值时，波被相消干涉。

*(a) 波 $y_1(x,t)$ 和 $y_2(x,t)$ 的传播方向是什么？ (b) 在 $t = (\pi/50)$ s 时，两个波在什么位置 x 发生相长干涉，$|y_s|$ 对应的值是多少？ (c) 在 $t = (\pi/50)$ s 处，两个波在 x 的什么位置发生相消干涉，对应的 $|y_s|$ 值是多少？

1.9 求出正弦波在负 x 方向上沿弦传播的 y(x,t) 表达式，假设 $y_{max} = 40$ cm，$\lambda = 30$ cm，$f = 10$ Hz，并且

(a) $y(x, 0) = 0$ 在 $x = 0$ (b) $y(x, 0) = 0$ 在 $x = 3.75$ cm *

1.10 在弦上产生正弦波的振荡器在 50 秒内完成 20 次振动。观察到波峰在 5 s 内沿着弦传播了 2.8 m 的距离。波长是多少？

1.11 弦的垂直位移由调和函数给出：

$$y(x,t) = 2\cos(16\pi t - 20\pi x) \qquad \text{(m)},$$

其中 x 是沿绳索的水平距离（以米为单位）。假设一个微小的颗粒附着在 $x = 5$ 厘米处的绳子上。获得粒子垂直速度随时间变化的表达式。

*1.12 给定两个波，其特征为

$$y_1(t) = 3\cos\omega t,$$
$$y_2(t) = 3\sin(\omega t + 60^\circ),$$

$y_2(t)$ 超前还是滞后 $y_1(t)$ 以及相位角是多少？

1.13 在传输线上传播的电磁波的电压由下式给出

$$v(z,t) = 5e^{-\alpha z}\sin(4\pi \times 10^9 t - 20\pi z) \qquad \text{(V)},$$

其中 z 是距发电机的距离（以米为单位）。

(a) 求出波的频率、波长和相速度。 (b) 在 $z = 2$ m 处，测得波的振幅为 2 V。求出 α。

*1.14 某电磁波在海水中传播，在10m深度处观测到振幅为98.02（V/m），在100m深度处振幅为81.87（V/m）。海水的衰减常数是多少？

1.15 穿过雾气的激光束在距离激光枪 2 m 处观察到强度为 1 (μW/m^2)，在距离激光枪 3 m 处强度为 0.2 (μW/m^2)。假定电磁波的强度与其电场幅度的平方成正比，求雾的衰减常数 α。

1.16 在模块 1.2 中，设置 $f = 1$ Hz、$\lambda = 2$ cm 和 $\alpha = 0.5$ Np/cm。估计蓝色波在第一个最大值（$x = 0$ 处）和第二个最大值（$x = 2$ cm 处）的振幅 y(x)。计算比率 $y(2\text{ cm})/y(0)$。给定 f 、 λ 和 α 的值，结果是否符合您的预期？

第 1-5 节：复数

1.17 复数 z_1 和 z_2 给出为

$$z_1 = 3 - j2,$$
$$z_2 = -4 + j3.$$

(a) 以极坐标形式表示 z_1 和 z_2。

(b) 首先应用方程求出 $|z_1|$。 （1.41），然后应用方程。（1.43）。

*(c) 确定极坐标形式的乘积 $z_1 z_2$。

(d) 确定极坐标形式的比率 z_1/z_2。

(e) 确定极坐标形式的 z_1^3。

1.18 计算下列每个复数并用直角形式表示结果：

(a) $z_1 = 8e^{j\pi/3}$

***(b)** $z_2 = \sqrt{3}\, e^{j3\pi/4}$

(c) $z_3 = 2e^{-j\pi/2}$

(d) $z_4 = j^3$

(e) $z_5 = j^{-4}$

(f) $z_6 = (1-j)^3$

(g) $z_7 = (1-j)^{1/2}$

1.19 Complex numbers z_1 and z_2 are given by

$$z_1 = -3 + j2,$$
$$z_2 = 1 - j2.$$

Determine (a) $z_1 z_2$, (b) z_1/z_2^*, (c) z_1^2, and (d) $z_1 z_1^*$ all in polar form.

1.20 If $z = -2 + j4$, determine the following quantities in polar form:

(a) $1/z$
(b) z^3
*__(c)__ $|z|^2$
(d) $\Im\mathrm{m}\{z\}$
(e) $\Im\mathrm{m}\{z^*\}$

1.21 Find complex numbers $t = z_1 + z_2$ and $s = z_1 - z_2$, both in polar form, for each of the following pairs:

(a) $z_1 = 2 + j3$ and $z_2 = 1 - j2$
(b) $z_1 = 3$ and $z_2 = -j3$
(c) $z_1 = 3\angle 30°$ and $z_2 = 3\angle -30°$
*__(d)__ $z_1 = 3\angle 30°$ and $z_2 = 3\angle -150°$

1.22 Complex numbers z_1 and z_2 are given by

$$z_1 = 5\angle -60°,$$
$$z_2 = 4\angle 45°.$$

(a) Determine the product $z_1 z_2$ in polar form.
(b) Determine the product $z_1 z_2^*$ in polar form.
(c) Determine the ratio z_1/z_2 in polar form.
(d) Determine the ratio z_1^*/z_2^* in polar form.
(e) Determine $\sqrt{z_1}$ in polar form.

*__1.23__ If $z = 3 - j5$, find the value of $\ln(z)$.

1.24 If $z = 3e^{j\pi/6}$, find the value of e^z.

1.25 If $z = 3 - j4$, find the value of e^z.

Section 1-6: Phasors

*__1.26__ A voltage source given by

$$v_s(t) = 25\cos(2\pi \times 10^3 t - 30°) \quad \text{(V)}$$

is connected to a series RC load as shown in **Fig. 1-20**. If $R = 1\ \text{M}\Omega$ and $C = 200$ pF, obtain an expression for the voltage across the capacitor, $v_c(t)$.

1.27 Find the instantaneous time sinusoidal functions corresponding to the following phasors:

(a) $\tilde{V} = -5e^{j\pi/3}$ (V)
(b) $\tilde{V} = j6e^{-j\pi/4}$ (V)
(c) $\tilde{I} = (6 + j8)$ (A)
*__(d)__ $\tilde{I} = -3 + j2$ (A)
(e) $\tilde{I} = j$ (A)
(f) $\tilde{I} = 2e^{j\pi/6}$ (A)

1.28 Find the phasors of the following time functions:

(a) $v(t) = 9\cos(\omega t - \pi/3)$ (V)
(b) $v(t) = 12\sin(\omega t + \pi/4)$ (V)
(c) $i(x,t) = 5e^{-3x}\sin(\omega t + \pi/6)$ (A)
*__(d)__ $i(t) = -2\cos(\omega t + 3\pi/4)$ (A)
(e) $i(t) = 4\sin(\omega t + \pi/3) + 3\cos(\omega t - \pi/6)$ (A)

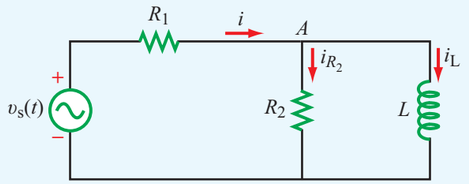
1.29 A series RLC circuit is connected to a generator with a voltage $v_s(t) = V_0\cos(\omega t + \pi/3)$ (V).

(a) Write the voltage loop equation in terms of the current $i(t)$, R, L, C, and $v_s(t)$.
(b) Obtain the corresponding phasor-domain equation.
(c) Solve the equation to obtain an expression for the phasor current $\tilde{I}$.

1.30 The voltage source of the circuit shown in **Fig. P1.30** is given by

$$v_s(t) = 25\cos(4 \times 10^4 t - 45°) \quad \text{(V)}.$$

Obtain an expression for the current flowing through the inductor, $i_L(t)$.



$R_1 = 20\ \Omega,\ R_2 = 30\ \Omega,\ L = 0.4$ mH

Figure P1.30 Circuit for Problem 1.30.

1.31 Repeat Problem 1.30, but this time determine current $i_{R_2}(t)$.

1.19 复数 z_1 和 z_2 为

$$z_1 = -3 + j2,$$
$$z_2 = 1 - j2.$$

用极坐标形式表示：(a) z_1z_2；(b) z_1/z_2^*；(c) z_1^2；(d) $z_1z_1^*$。星号表示复共轭。

1.20 若 z = −2 + j4，求下列各量的极坐标形式：

(a) $1/z$
(b) z^3
*__(c)__ $|z|^2$
(d) $\Im\mathrm{m}\{z\}$
(e) $\Im\mathrm{m}\{z^*\}$

1.21 对下列各组复数，求 t = z_1 + z_2 和 s = z_1 − z_2，并用极坐标形式表示：
(a) z_1 = 2 + j3，z_2 = 1 − j2
(b) z_1 = 3，z_2 = −j3
(c) $z_1 = 3\angle 30°$，$z_2 = 3\angle -30°$
*(d) $z_1 = 3\angle 30°$，$z_2 = 3\angle -150°$

1.22 复数 z_1 和 z_2 为

$$z_1 = 5\angle -60°,$$
$$z_2 = 4\angle 45°.$$

(a) 用极坐标形式表示乘积 z_1z_2。
(b) 用极坐标形式表示乘积 $z_1z_2^*$。
(c) 用极坐标形式表示比值 z_1/z_2。
(d) 用极坐标形式表示比值 z_1^*/z_2^*。
(e) 用极坐标形式表示 $\sqrt{z_1}$。

*1.23 若 z = 3 − j5，求 ln(z)。

1.24 若 $z = 3e^{j\pi/6}$，求 e^z。

1.25 若 z = 3 − j4，求 e^z。

相量（第 1-7 节）

*1.26 电压源

$$v_s(t) = 25\cos(2\pi \times 10^3 t - 30°) \quad \text{(V)}$$

接至图 1-20 所示的串联 RC 负载。若 R = 1 MΩ、C = 200 pF，求电容器两端电压 $v_c(t)$ 的表达式。

1.27 求与下列相量对应的瞬时正弦时间函数：

(a) $\tilde{V} = -5e^{j\pi/3}$ (V)
(b) $\tilde{V} = j6e^{-j\pi/4}$ (V)
(c) $\tilde{I} = (6 + j8)$ (A)
*__(d)__ $\tilde{I} = -3 + j2$ (A)
(e) $\tilde{I} = j$ (A)
(f) $\tilde{I} = 2e^{j\pi/6}$ (A)

1.28 求下列时间函数对应的相量：

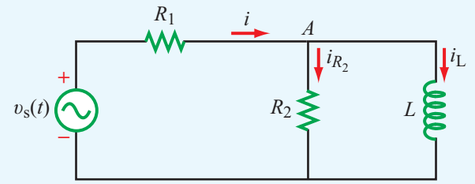
(a) $v(t) = 9\cos(\omega t - \pi/3)$ (V)
(b) $v(t) = 12\sin(\omega t + \pi/4)$ (V)
(c) $i(x,t) = 5e^{-3x}\sin(\omega t + \pi/6)$ (A)
*__(d)__ $i(t) = -2\cos(\omega t + 3\pi/4)$ (A)
(e) $i(t) = 4\sin(\omega t + \pi/3) + 3\cos(\omega t - \pi/6)$ (A)

1.29 串联 RLC 电路接至电压为 $v_s(t) = V_0\cos(\omega t + \pi/3)$ (V) 的电源。
(a) 用电流 i(t)、R、L、C 和 $v_s(t)$ 写出回路电压方程。
(b) 写出对应的相量域方程。
(c) 解此方程，求出电流相量。

1.30 图 P1.30 所示电路的电压源为

$$v_s(t) = 25\cos(4 \times 10^4 t - 45°) \quad \text{(V)}.$$

求流过电感器的电流 $i_L(t)$ 的表达式。



$R_1 = 20\ \Omega,\ R_2 = 30\ \Omega,\ L = 0.4$ mH

图 P1.30 习题 1.30 的电路。

1.31 对习题 1.30 的电路，求电流 $i_{R2}(t)$。

校注：本页原书的“Section 1-6: Phasors”与正文节号不一致；相量对应第 1-7 节。